



Analytics & Data Science Portfolio

Strategic Impact Through Data-Driven Solutions



EXECUTIVE SUMMARY

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Profile

Strategic CX Data Analytics Manager | Data Science Leader | Analytics Trainer with **10+ years** at Telecom Egypt (WE)—progressing from founding analyst to leading **7+ analysts** and **100+ stakeholders** across **12+ departments**. Built and scaled enterprise analytics capabilities serving **22M+ customers**: **McKinsey-adopted personas (replacing their internal models)**, enterprise churn prediction achieving **74%+ recall** across **10M** customers, **R²/Adjusted R² 0.45–0.99** CX driver analysis.

Program-led the **8-workstream** CX Culture initiative and pioneered the organization's **first CX Culture Assessment**—establishing a quantified baseline and surfacing **critical gaps** now steering transformation priorities.

I bridge enterprise strategy, advanced analytics & data science, and organizational capability building.

- **Strategic Orchestration:** Delivered multiple **high-impact** enterprise programs achieving **TNPS +22pts**, chat dropped sessions **-68%**, **200K+** detractors handled through closed-loop recovery program (pilot: **61.4%** converted to Promoters), report turnaround **-42%**, and **30+** executive dashboards.
- **Advanced Analytics & Data Science:** Technical credibility validated by Kaggle **Top 30% ranking**. Designed and deployed **production-ready machine learning solutions** in partnership with IT teams spanning regression, classification, clustering, time series forecasting, and experimentation frameworks.
- **Capability Building:** Delivered training that impacted **250+** professionals (**540+** hours, **98%** satisfaction) across Egypt & Saudi Arabia—translating complex analytics into "**Decision Intelligence**" that informs enterprise strategy and drives sustained commercial advantage.

Leadership & Technical Breadth

- **Industries:** Telecommunications, Retail, and Financial Services & Insurance (DS Projects).
- **CX Expertise:** CXM • CX Metrics (NPS/TNPS/CSAT/CES/CLV) • Journey Analytics • Touchpoints Analytics • Personas • VoC/VoE • Churn • Driver Analysis • KPI Governance.
- **Leadership:** Team Management (7+ analysts) • Program Management • Cross-Functional Orchestration • C-Suite Communication • Training & Mentorship.
- **ML Methods:** Regression • Classification • Clustering (K-Means, Neural Gas) • Time Series (ETS, ARIMA) • A/B Testing.
- **Technical Stack:** Python • SQL • Power BI (DAX, Power Query) • Excel • Alteryx • KNIME • Tableau.



CONFIDENTIALITY NOTE

- **To honor non-disclosure agreements and protect proprietary business data**, certain internal organizational metrics (baselines, gaps, and diagnostic scores) are marked **[CONFIDENTIAL]**.
- Methodology details, statistical validation (Cronbach's α , Confidence Levels), scale metrics, and externally validated achievements **remain fully disclosed**.
- Detailed metrics available for discussion in interview settings under appropriate confidentiality agreements.



Certificates

- **Microsoft Certified PL-300: Data Analyst Associate** | Microsoft | **2021-2027**
- **Foundation of Digital Marketing** | Udacity | **2025**
- **AWS Educate Machine Learning Foundations** | AWS | **2025**
- **McKinsey Leadership Forward Program** | McKinsey & Company | **2024**
- **Probability and Statistics** | University of London | **2022**
- **Predictive Analytics for Business** | Udacity | **2021**
- **AWS Machine Learning** | Udacity | **2021**
- **30 Days of Machine Learning** | Kaggle Challenge | **2021**
- **Data Analysis Professional Nanodegree** | Udacity | **2020**
- **Target Audience Training: Identify Your Ideal Customer** | Udemy | **2020**
- **User Experience Tools** | Udemy | **2019**
- **Scrum Fundamentals Certified** | SFC | **2019**

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Strategic Corporate Experience & CX Culture Transformation

1. Data-Driven Customer Personas Development

Organization: Telecom Egypt (WE) | **Timeline:** 2019-2020 (later adopted in 2022)

Role: Senior Data Analyst & Project Lead

Scale: 6.3M+ customers • 2.4M+ transactions • 1.5K+ retail branches • Mixed quantitative + qualitative inputs

Business Problem & Objective

- **Situation:** WE lacked a unified, data-driven view of its **6.3M+ fixed broadband customers**, resulting in generic campaigns, inconsistent targeting across retail and digital channels, and missed personalization opportunities; segmentation relied mainly on demographics and ARPU.
- **Task:** Build actionable personas grounded in **behavior + demographics + primary research** to enable better CX and commercial targeting.

My Role & Approach

- Led end-to-end persona development initiative from methodology design to enterprise rollout; adapted **CRISP-DM** for persona development.
- Integrated CRM and transactional data with primary research (customer profiling survey + focus groups) to build a representative analytical sample (~26K customers; 2.4M+ transactions; **99% CI, 1% margin of error**).
- Engineered **15+ behavioral/demographic features** and conducted data cleaning + **EDA** to prepare for segmentation/clustering.
- **Built** and validated **K-means clustering solution**, defining **5 personas** (silhouette scores **0.41-0.47**).
- Developed interactive **Power BI** dashboards and delivered executive readouts/workshops to drive adoption across Marketing, Sales, Product, Retail, and Customer Care.

Deliverables

- Persona framework (definitions, sizing, key behaviors, needs, and channel preferences).
- Interactive Power BI persona dashboards for non-technical stakeholders.
- Persona enablement sessions/playbook to operationalize usage in CX and targeting.

Key Outcomes & Impact

- **McKinsey Adoption:** Personas officially adopted by McKinsey as the WE standard during their 2022 SAHM project engagement—**replacing McKinsey's internal models**.
- **5 data-driven personas** built using behavioral analytics + primary research; used as a shared customer language across CX, Marketing, Retail and Customer Care teams.

Tools & Techniques

SQL • Excel Power Query (M) • Access • Power BI (data modeling, DAX) • Python/Alteryx/KNIME • EDA • Feature engineering • K-means clustering • Workshop facilitation

Skills Demonstrated

Customer Segmentation • Persona Development • Unsupervised ML (Clustering) • Feature Engineering • Dashboarding • Stakeholder Management • C-Suite Communication • ETL • Statistical Sampling & Validation • Research Methodology Design • Enterprise-Level Influence.

2. Data-Driven Customer Personas Development (Enterprise Refresh: FBB + Mobile)

Organization: Telecom Egypt (WE) | **Timeline:** 2023-Present (Phase 4 ongoing; target completion Q4-2026)

Role: CX Data Analytics Manager (Strategic Orchestrator & Consultant)

Scale: 10M+ FBB customers • 12M+ Mobile customers • **DWH-scale** clustering on full population • 16 factors (FBB) / 30 factors (Mobile) • Cross-functional stakeholders (IT Data Modeling/BI, CVM & Retention, Customer Care, Marketing, CX)

Business Problem & Objective

- **Situation:** Following the success of 2019 personas, the organization needed to evolve its customer segmentation approach—covering both FBB (10M+) and Mobile (12M+) customers at full population scale. The 2019 personas became outdated and did not cover Mobile customers. WE initiated a new enterprise-wide persona program to leverage **IT DWH big-data** capabilities and align **Commercial (CVM/Retention)** and **Customer Care** decision-making on a unified, data-driven segmentation approach.
- **Task:** Orchestrate an enterprise-wide persona initiative to build actionable, data-driven personas grounded in **behavior + demographics** (and upcoming **primary research**), ensuring early **stakeholder involvement** and enterprise-wide **usability and adoption across business** teams.

My Role & Approach

- **Strategic Orchestration**—aligned IT (data modeling/BI) and business stakeholders (CVM, Retention, Customer Care, Marketing, CX) early on requirements, use cases, and clustering factors.
- **Factor Architecture:** Guided the selection of 16 behavioral/demographic factors for **FBB** personas; 30 factors for **Mobile** personas—balancing analytical rigor with business interpretability.
- **Phased Governance:** Guided phased delivery and governance across Phase 1 (clustering factors selection), Phase 2 (DWH-scale K-means clustering), Phase 3 (clustering results dashboard), Phase 4 (business validation + profiling survey preparation; in progress).
- **Technical Advisory:** Ensured clustering logic and factor definitions were business-relevant and technically implementable (**bridging business needs with DWH/data-model constraints**).
- Led **model validation discussions and interpretation** (FBB evaluated; Mobile/GSM validation pending).
- **Validation Leadership:** Identified early refinement opportunities from validation outputs (e.g., potential feature scaling/weighting adjustments; optional secondary algorithm testing if needed to reduce mild overlaps).

Deliverables

- Stakeholder-aligned clustering factors documentation (16 factors for FBB; 30 factors for Mobile).
- Enterprise-scale K-means clustering execution outputs (separate FBB and Mobile; final personas TBD post-validation).
- Clustering-results Power BI dashboard for validation and business interpretation (audience: CX, CVM, Retention, Marketing, Customer Care).
- Phase 4 validation framework and profiling survey preparation (in progress).

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Key Outcomes & Impact

- **Phase progress: Completed Phase 1-3** (factor definition, clustering execution, and dashboard delivery); currently executing Phase 4 business validation and preparing profiling survey.
- **Early validation insight (FBB only):** clustering quality indicators show strong segmentation structure—compactness **0.06-0.19**, distinctiveness **1.3-1.95**, and approximated average silhouette **~0.92**; a subset of clusters shows mild overlap indicating selective refinement opportunities (**Mobile/GSM not evaluated yet**).

Tools & Techniques

K-means clustering • Feature/factor design (16 FBB / 30 Mobile) • Segmentation validation (compactness, distinctiveness, silhouette) • Power BI dashboard review/validation • Cross-functional workshops & governance • Enterprise DWH execution.

Skills Demonstrated

Cross-Functional Orchestration • Strategic Advisory • Analytics consulting/advisory • Stakeholder Alignment • ML Clustering (K-means) • Clustering Validation & Interpretation • Enterprise-Scale Analytics • Change Management • C-Suite Communication • Methodology Design • Multi-Team Coordination

3. Platinum Customer Segment: Strategic Experience Transformation

Organization: Telecom Egypt (WE) | **Timeline:** 2024–2025 (Discontinued)

Role: CX Data Analytics Manager (Project Orchestrator & Strategic Consultant — CEO-Directed Initiative)

Scale: [CONFIDENTIAL] Platinum High-Value Customers (Targeted Optimization to [CONFIDENTIAL]) • Multi-Channel Pilot Design • [CONFIDENTIAL] exchanges • 8+ Cross-Functional Stakeholder Divisions

Business Problem & Objective

- **Situation:** The organization sought to differentiate service for its "Platinum" high-value segment to drive retention in a competitive market. However, the existing journey was standardized, and operational units (Regions, Retail, IT, Customer Care) lacked alignment on resource requirements and service levels needed to deliver premium treatment.
- **Task:** Orchestrate a comprehensive, **data-backed multi-channel service strategy** defining aggressive yet operationally sustainable SLAs—balancing "Platinum Treatment" expectations with frontline capacity constraints—across **8+ stakeholder teams**.

My Role & Approach

- **Strategic Orchestration:** Served as central **bridge** between **CEO vision and 8+ cross-functional teams** (Strategy, IT, Commercial, Regions, Customer Care, Retail), navigating competing requirements to build a unified operational roadmap.
- **Analytical Advisory:** Partnered with **IT DWH** team to analyze behavioral profiles of [CONFIDENTIAL] Platinum customers; recommended strategic reduction to [CONFIDENTIAL] customers to ensure proposed high-touch SLAs (e.g., [CONFIDENTIAL]-hour resolution) remained operationally sustainable.
- **Business Insight & Stakeholder Alignment:** Uncovered that [CONFIDENTIAL] of exchanges ([CONFIDENTIAL] out of [CONFIDENTIAL]) serve fewer than [CONFIDENTIAL] Platinum customers each. This single data point helped align operational and commercial teams—removing communication barriers and enabling rapid consensus on resource allocation.
- **SLA Architecture & Negotiation:** Led definition and negotiation of 15+ "Platinum-Grade" service standards across channels:
 - Customer Care: [CONFIDENTIAL] SLA handled by empowered senior agents (*answer [CONFIDENTIAL] of Customers within [CONFIDENTIAL] seconds*).
 - Technical Support: [CONFIDENTIAL] resolution within [CONFIDENTIAL] hours + free technical home visits.
 - Retail: "First-to-Serve" priority queuing + dedicated logistics windows.
- **Technical Design:** Collaborated with IT to architect "Platinum Class" **identification tags within CRM**, enabling seamless customer recognition across 1,500+ branches and digital channels.

Deliverables

- **Strategic Pilot Framework:** Detailed design for two-exchange pilot ([CONFIDENTIAL] and [CONFIDENTIAL]) testing specialized technical support feasibility.
- **Multi-Channel SLA Matrix:** Governance document defining 15+ service standards across billing, logistics, and technical complaints.
- **Resource Optimization Blueprint:** Capacity plan identifying required technician/agent headcounts to maintain premium service without compromising standard operations.

Key Outcomes & Impact

- **Executive Alignment:** Secured alignment/sign-off on pilot design, including frontline compensation and refund empowerment model.
- **Stakeholder Consensus:** Distribution insight ([CONFIDENTIAL] exchanges with <[CONFIDENTIAL] Platinum customers) removed communication barriers and enabled rapid operational alignment.
- **Data-Driven Governance:** Defined a bi-annual customer list refresh protocol to stabilize resource planning and preventing segment "scope creep".
- **Framework Delivered:** Completed fully documented service model standardizing "Common Customer Language" for high-value treatment—ready for execution despite leadership transition.
- **Operational Sustainability:** Strategic segment reduction ([CONFIDENTIAL] →[CONFIDENTIAL]) enabled realistic SLAs balancing premium service with frontline capacity.

Note: Project discontinued in 2025 following CEO transition and shifting stakeholder priorities; the pilot framework and SLA artifacts were completed and handed over.

Tools & Techniques

DWH Behavioral Analysis • Distribution Analysis • SLA Design & Negotiation • Gap Analysis • Capacity Planning • Stakeholder Orchestration • Service Design • Business Requirements Documentation (BRD) • CRM Integration Architecture

Skills Demonstrated

Strategic Advising • Cross-Functional Orchestration • CEO-Level Stakeholder Management • Operational Strategy • Change Management • Conflict Resolution • Resource Optimization • Service Level Agreement (SLA) Design • Data-Driven Decision Making • Insight Translation • Stakeholder Alignment • Navigating Organizational Transition

4. CX Culture Program (Enterprise Transformation)

Organization: Telecom Egypt (WE) | **Timeline:** May 2025–Present (Pilot completed Mar 2026; enterprise rollout pending)

Role: CX Data Analytics Manager (Program Manager — Executive-Sponsored Initiative)

Scale: 8 integrated projects • 6,000+ Direct Sales employees (Pilot) • 60,000 employees (full rollout target) • **Agile/Scrum** governance • **Cross-functional** delivery (CX, Direct Sales, UX/UI, HR, Media)

Business Problem & Objective

- **Situation:** The organization needed to shift from a traditional operational mindset to a customer-centric culture across customer-facing and non-customer-facing employees. There was no unified baseline maturity measurement, reinforcement engine, or engagement mechanism to embed CX values and sustain culture change at scale.
- **Task:** Design and lead an enterprise-wide CX Culture Change roadmap comprising 8 integrated projects to define baseline CX culture level, create scalable internal CX content, gamified engagement mechanisms, and tracking systems to drive sustained cultural transformation with measurable employee engagement—tracking progress through CX Culture (CC) Scores and CX KPIs (e.g., NPS/CSAT).

My Role & Approach

- **Program Leadership:** Defined program vision, success criteria, managing dependencies, and roadmap for 8 integrated workstreams; secured sponsorship from CX Sr. Director and Direct Sales Director.
- **Portfolio Orchestration:** Coordinated dependencies across 8 projects: CX Culture Assessment, CX Ambassadors, CX Tips, CX Talks (Podcast), CX Awareness Videos, CX Dynamic Signature, Life at WE, and Rewards & Incentives.
- **Pilot Design & Execution:** Defined the Direct Sales pilot scope (6,000+ employees) and executed a Phase-1 bundle: CX Culture Assessment, CX Tips, CX Talks (Podcast), CX Awareness Videos, Rewards & Incentives.
- **Content Strategy:** Built unified content theme framework; oversaw creation and validation of 35 multimedia assets; collaborated with UX/UI team on visual design and usability.
- **Engagement mechanics:** Designed and implemented a points-based, multi-weighted scoring system integrating survey participation, quiz accuracy, and comment engagement—ensuring winner selection based on a holistic 360° engagement profile.
- **Governance & cadence:** Managed end-to-end delivery using Agile Scrum framework with weekly steering cadence; addressed recurring team dynamics to maintain program stability.

Deliverables

- Program **governance framework** with 8-project roadmap and success criteria.
- **Content library:** 35 multimedia assets (24 CX tips, 6 awareness videos, 5 podcast episodes) designed for long-term cultural onboarding.
- **Points-based engagement system** integrating survey completion, quizzes, and comments with transparent leaderboard logic.
- **Engagement tracking dashboard** monitoring participation, knowledge validation, and cultural sentiment.
- **Winner selection framework:** Top [CONFIDENTIAL] employees identified via combined leaderboard (points range: [CONFIDENTIAL] to [CONFIDENTIAL]).

Key Outcomes & Impact

- **Pilot completion:** Direct Sales pilot delivered (enterprise rollout pending with HR).
- **Survey participation:** **1,064** employees completed the CX Culture Assessment survey ([CONFIDENTIAL] of 6,000+ pilot population; **95%** CL, **3%** margin of error).
- **Knowledge engagement (quizzes):** [CONFIDENTIAL] quiz submissions from [CONFIDENTIAL] unique employees ([CONFIDENTIAL] of survey respondents; avg [CONFIDENTIAL] submissions per quiz participant).
- **Interaction engagement (comments):** [CONFIDENTIAL] comments from [CONFIDENTIAL] unique employees ([CONFIDENTIAL] of survey respondents; avg [CONFIDENTIAL] comments per commenter).
- **Total engagement actions:** [CONFIDENTIAL] combined quizzes + comments actions across pilot content.
- **Leaderboard (base = survey respondents):** Leaderboard population is the same **1,064** employees; scoring combines survey + quizzes + comments points.
- **Recognition readiness:** Top score [CONFIDENTIAL] points ([CONFIDENTIAL] employees tied at **Rank #1**); **Top-[CONFIDENTIAL]** recognition threshold reached [CONFIDENTIAL] points (Rank #[CONFIDENTIAL]); points distribution spanned [CONFIDENTIAL] ranks (including employees with 0 points).
- **Content signals (examples):** Highest quiz response Tip-1 ([CONFIDENTIAL]); highest comment volume Culture Assessment ([CONFIDENTIAL]).

Tools & Techniques

Agile (Scrum) • Program Governance • Content Strategy • Gamification Design • Multi-Weighted Scoring Logic • MS Forms • Power BI Dashboard • Stakeholder Orchestration • Change Management • Engagement Analytics

Skills Demonstrated

Program Management • Strategic Leadership • Cross-Functional Orchestration • Change Management • Content Strategy • Gamification & Behavioral Design • Agile/Scrum Governance • Executive Stakeholder Management • Cultural Transformation • Pilot Design & Execution • Engagement Measurement • Team Stability Management

5. CX Culture Assessment (Benchmarking Organizational Culture Level)

Organization: Telecom Egypt (WE) | **Timeline:** Nov-2025–Mar-2026 (Part of CX Culture Change Program – baseline)

Role: CX Data Analytics Manager (Project Owner & Methodologist)

Scale: Pilot population: **6,000+** Direct Sales employees • Survey sample: **1,064** responses (**95%** CL, **3%** MoE)
• 5-Dimension Maturity Framework (**25** Attributes) • CC Score framework • Power BI Dashboard

Business Problem & Objective

- **Situation:** The organization lacked a quantified baseline of CX culture maturity. No framework existed to measure employee perceptions, identify cultural gaps across departments/levels, or track transformation progress. Leadership needed a data-driven foundation to guide and prioritize targeted interventions and measure program ROI.
- **Task:** Design and execute a comprehensive CX Culture Assessment—developing a reusable survey instrument, establishing a CC Score methodology, and delivering actionable insights to identify strengths and weaknesses, informing the broader CX Culture Change Program either by steering the 8-project CX Culture Change roadmap or initiating new programs/projects to enable future ROI measurement.

My Role & Approach

- **Survey Methodology Design:** Designed and implemented a comprehensive CX Culture survey while adapting Swisscom/Lucerne Institute established CC Score framework for the Egyptian telecom context; refined through 4 iterative pilots before full deployment.
- **Instrument Development:** Designed comprehensive survey covering 5 dimensions comprising the Customer Centricity Score framework (Leadership, Collaboration, Implementation, Values & Beliefs, CX Knowledge Check)—each dimension with 5 measurable attributes.
- **Defined Scoring Logic (CC Score Framework):** Defined calculation logic as (Count of 4&5 ratings – Count of 1&2 ratings) ÷ Total responses; excluded “Don’t know/Can’t understand” (6 and 7 rates).
- **Anonymous Participation Methodology Innovation:** Designed methodology enabling real anonymous responses while tracking eligibility for points system—increasing sample size and capturing authentic Voice of Employee (VOE) without fear of consequences.
- **Sampling Framework:** Ensured statistical rigor by defining statistically representative sample (1,064 respondents; 95% confidence level, 3% margin of error) from 6,000+ Direct Sales population.
- **Reliability Validation:** Validated the instrument’s internal consistency using Cronbach’s Alpha, achieving high reliability scores; Overall $\alpha=0.92$ (Likert-scale culture items), Leadership $\alpha=0.67$, Collaboration $\alpha=0.62$, Implementation $\alpha=0.83$, Values & Beliefs $\alpha=0.82$.
- **Dashboard Development:** Built Power BI dashboard visualizing CC Score by dimension, attribute-level radar charts, and perception-vs-reality gap analysis breakdowns for leadership review and program prioritization.
- **Insight Translation:** Identified critical gaps and priority intervention areas; mapped findings to specific program projects.

Deliverables

- **Survey instrument ready** for reuse (survey-design ready to assess CX culture in other organizations).
- **CC Score scoring methodology**, calculation logic, reporting definition, and governance documentation.
- **Power BI dashboard** with dimension-level visualizations and gap analysis.
- **Assessment report** with baseline scores, key themes, and actionable recommendations.
- **Potential CX Ambassador** candidate identification based on survey responses.
- Insight pack to **inform culture program prioritization** (e.g., incentives, ambassadors focus areas, content themes, and potential new initiatives).

Key Outcomes & Impact

- **Quantified Culture Level Baseline:** Established the first CC Score baseline initial result (Actual CC Score [CONFIDENTIAL] and Overall CC Score: [CONFIDENTIAL]), providing a rigorous benchmark for future cultural ROI.
- **The "Experience Gap" Discovery:** Uncovered a significant [CONFIDENTIAL] Perception-vs-Reality Gap, identifying that while leadership prioritized CX ([CONFIDENTIAL]), the reward systems were severely lacking ([CONFIDENTIAL]), indicating disconnect between perceived and lived CX culture.
- **Pillar Diagnostic:**
 - **Leadership ([CONFIDENTIAL]):** Leadership seen as making CX a priority, but there is a massive opportunity in rewarding CX behaviors (Incentives [CONFIDENTIAL]), validated urgency of Rewards & Incentives project.
 - **Collaboration ([CONFIDENTIAL]):** Identified as the lowest-performing pillar, validating the urgent need for the CX Ambassadors project to break operational silos (Alignment [CONFIDENTIAL]).
 - **Implementation ([CONFIDENTIAL]):** Highlighted strong data usage ([CONFIDENTIAL]) but low individual ownership (Personal Responsiveness [CONFIDENTIAL]), leading to new employee empowerment initiatives.
 - **Values & Beliefs ([CONFIDENTIAL]):** Strong shared responsibility ([CONFIDENTIAL]) contrasted with low employee support (Employee Experience [CONFIDENTIAL]), directly informing the Life at WE program.
- Insights were used to guide program focus and reinforce urgency around Rewards & Incentives and cross-functional alignment interventions (program execution continues).

Tools & Techniques

Survey design • Statistical Sampling approach • Cronbach's Alpha Validation • MS Forms • Power BI Desktop • Scoring methodology design • Diagnostic analytics • Culture maturity benchmarking approach • Agile/Scrum Governance Alignment

Skills Demonstrated

Survey Design & Methodology • Assessment Framework Development • KPI/Scoring Framework Design • Statistical Sampling • Anonymity Design Innovation • Power BI Dashboarding & executive reporting • Stakeholder influence • Data-Driven Insight Translation • Culture Analytics • Benchmarking Adaptation • Pilot Iteration & Refinement

6. Churn Analysis (Descriptive & Diagnostic Analytics for Voluntary Churn)

Organization: Telecom Egypt (WE) | **Timeline:** 2019 (Study Period: Jan 2018–May 2019, Data Snapshot: May-2019, Refined through 2021)

Role: CX Data Analyst (Solo Project Lead)

Scale: 10K subscribers (random sample from survey responses) • Monthly churn rate [CONFIDENTIAL] • Estimated ~EGP [CONFIDENTIAL] avg monthly loss (derived from loss metric; not audited finance) • Error margin <3% • Power BI dashboard (8 pages) + framework + report • [CONFIDENTIAL] early failure contribution (first 30 days)

Business Problem & Objective

- **Situation:** WE needed a data-driven understanding of voluntary churn to reduce cancellations. Stakeholders lacked a unified view of who churns, when churn happens, key churn drivers, and how to translate findings into actionable retention priorities—especially linking churn to CX signals (NPS) and operational friction (tickets/repeats).
- **Task:** Perform a comprehensive descriptive and diagnostic churn analysis on voluntary churn customers—answering four critical research questions: Who is leaving? When are they leaving? Why are they leaving? How to retain them? to recommend specific, data-backed actions for churn reduction.

My Role & Approach

- **Solo Project Leadership:** Owned end-to-end analysis from methodology design to final recommendations; first churn analysis task of its kind for me; repeated the project 4 times, including a 2021 iteration collaborating with Customer Journey team on focus calls.
- **Methodology Design:** Defined data collection and analysis process; established NPS methodology differentiating customers who never asked for cancellation (last survey) vs. those who did (last survey before cancellation)—enabling sentiment-churn correlation.
- **Metric Engineering:** Besides the commonly known KPIs (Churn Rate, ARPU, Actual Revenue, Estimated Loss, Package Price Variability “min/max/avg”), I architected and documented sophisticated churn KPIs and logic, AVG TT Complain (velocity to first friction point), Survey-to-Cancellation Velocity, and Average Actual Loss (Sum of lost MRR per churned customer).
- **Data Engineering:** Gathered and wrangled data from multiple sources (billing, tickets, cancellations, survey) using Excel Power Query and Access; ensured statistical rigor with <3% error margin (due to system update lags and data conversion gaps)
- **Exploratory Data Analysis (EDA):** Performed EDA to identify patterns across demographics, tenure, NPS segments, technical issues, and purchasing behavior which ended up with identified critical correlations between churn and technical performance, performing cluster analysis to segment attrition profiles.
- **Cluster Analysis:** Conducted clustering (with manager collaboration) to identify distinct churn segment profiles.
- **Dashboard Development:** Built 8-page interactive Power BI dashboard: Guidelines, Sample Statistics, Survey View, Churn View, Customer Age & Packages, Time to 1st Ticket, Geographic Map, Comprehensive Raw Data.
- **Recommendations:** Developed actionable retention recommendations and presented the findings to manager and CX team.

Deliverables

- **Churn analysis framework:** A ready methodology for ongoing diagnostics with clear definitions, metric logic, and analysis workflow.
- **8-page interactive Power BI dashboard** answering all research questions with drill-down capabilities.
- **Static report/presentation** summarizing findings and recommendations.

Key Outcomes & Impact

- **Reusable Methodology:** Produced a reusable churn diagnostic prototype that answered the core churn questions and became a baseline reference for later churn initiatives (repeated 4 times later).
- **NPS-Churn Correlation:** Answered the “**Who Question**” – Detractors' churn rate ([CONFIDENTIAL]) significantly higher than Passives + Promoters combined ([CONFIDENTIAL])
- **Early Churn Pattern:** Answered the “**When Question**” – [CONFIDENTIAL] of cancellation requests occurred within the first year of subscription.
- **Technical Issue Correlation:** Answered the “**Why Question**” – [CONFIDENTIAL] of customers requesting cancellation had a trouble ticket within the first 3 months; repeated/frequent problems were a major pattern.
- **Root Cause Identification:** Fostering the “**Why Question**” answer by declaring that **Early Failure (first 30 days)** represents [CONFIDENTIAL] of churn reasons; high volume of repeated/same problems identified.
- **Financial Impact:** Quantified [CONFIDENTIAL] EGP estimated average monthly revenue loss from voluntary churn (derived from loss metric; not audited finance).
- **Geographic Insight:** Highest churn volume in [CONFIDENTIAL]; most affected ARPU in [CONFIDENTIAL].
- **Strategic Recommendation:** Recommended focus on I-Complain journey and early-failure prevention —later aligned by consultant recommendations (McKinsey SAHM – 2022).

Tools & Techniques

Excel Power Query (M) • Microsoft Access • Power BI (ERD, Data Modeling, DAX) • PowerPoint • EDA • Cluster Analysis • NPS Segmentation • Statistical Sampling • Root Cause Analysis

Skills Demonstrated

Churn Analytics • Descriptive & Diagnostic Analytics • Metrics Design & Governance • Data Wrangling • EDA • Cluster Analysis • Power BI Dashboarding • NPS-Churn Correlation • Root Cause Analysis • Retention Strategy • First-Time Initiative Ownership

7. Churn Prediction ML Classification Model (FBB) — Proactive Retention Enablement

Organization: Telecom Egypt (WE) | **Timeline:** Mar 2022–Present (Model in production; commercial activation in progress)

Role: CX Sr. Data Analyst (Project Initiator & Designer) → CX Data Analytics Manager (Oversight & Strategic Advisor)

Scale: 10M FBB customers • **Daily** scoring cadence • **3** risk tiers Classification model (High/Medium/Low) • **74%+** Recall (churner class) • **CRM** Integration • **Cross-functional delivery** (IT Data Science, DWH BI, Commercial, Customer Care, Regions) • Power BI dashboard for CX & Commercial teams

Business Problem & Objective

- **Situation:** Building on the 2019 churn diagnostic insights, WE needed to shift from reactive churn analysis (why customers left) to proactive prevention (who and when will leave) by identifying at-risk customers early and enabling differentiated handling.
- **Task:** Initiate and orchestrate development of an enterprise-scale “Churn Prediction Supervised Machine Learning Classification Model”—defining requirements, features, and success criteria—to enable proactive, risk-based retention interventions across commercial, customer care, and field operations teams.

My Role & Approach

- **Initiative Ownership:** Initiated the churn prediction program (Mar 2022); authored comprehensive requirements document “Churn Analysis Project_V.1.0” (100+ data attributes across 8 categories: Churn Info, Demographics, Internet Usage Behavior, Problems & Journey Challenges, Purchasing Behavior, Subscriber Experience, Technical Insights, Subscriber Main Details).
- **Feature Engineering Architecture:** Defined detailed data mart specifications for IT; final model uses 5 key predictive factors: ([CONFIDENTIAL], [CONFIDENTIAL], [CONFIDENTIAL], [CONFIDENTIAL], and [CONFIDENTIAL]).
- **Cross-Functional Orchestration:** Aligned IT Data Science (model development), IT DWH BI (dashboard), Commercial (CVM/Retention), Customer Care, and Regional Technician teams on requirements, use cases, and adoption roadmap.
- **Dashboard Advisory:** Specified dashboard requirements for Power BI visualization; identified enhancement needs (histograms, filters, calculation corrections) for deeper commercial analysis.
- **Challenge Navigation:** Managed key obstacles including data availability, stakeholder adoption, and IT-business alignment.
- **Ongoing Oversight:** Continue to monitor model performance, dashboard accuracy, and commercial activation planning with stakeholders.

Deliverables

- **ML Feature Blueprint:** A comprehensive 100-variable data dictionary and calculation logic for churn modeling.
- **Production Risk Classification:** Daily scoring engine classifying the 10M+ base into 3 risk tiers (built by IT Data Science team).
- **Executive Power BI Dashboard:** Monitoring suite for CX and Commercial teams to track risk distribution (built by IT DWH/BI team).
- **CRM Integration Strategy:** Risk-based customer tagging (High/Medium/Low) to support differentiated handling design (commercial activation in progress).
- **Commercial activation roadmap:** In progress with CVM, Retention, and Customer Care.

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Key Outcomes & Impact

- **Enterprise Scale:** Deployed a churn prediction capability covering **10M FBB** customers.
- **Model Performance:** **74%+** Recall achieved (churner class); ongoing monitoring for refinement.
- **Daily Scoring:** Model runs daily, flagging customers by risk profile in CRM.
- **CRM Integration:** Each customer tagged (High/Medium/Low) for differentiated experience potential.
- **Cross-Team Alignment:** Successfully unified IT Data Science, DWH BI, Commercial, and Operations under shared churn prediction framework
- **Dashboard Evolution:** Identified and communicated enhancement requirements (histograms, filters, calculation corrections) to enable deeper commercial analysis.
- **Status:** Model in production; commercial action roadmap in development to translate predictions into retention interventions (business impact not yet measured).

Model developed by IT Data Science team based on my requirements, ongoing monitoring toward prediction accuracy. Commercial integration study in progress.

Tools & Techniques

Requirements Documentation • Data Mart Specification • Classification ML (model by IT) • Recall Metrics • Power BI Dashboard • CRM Integration • Cross-Functional Orchestration • Stakeholder Alignment • Predictive Analytics Governance

Skills Demonstrated

Predictive Analytics Strategy • Requirements Engineering • Analytics-to-ML Translation • Feature Architecture • Cross-Functional Orchestration • IT-Business Alignment • Data Product Ownership • CRM Integration Design • Stakeholder Management • Advisory Leadership • Challenge Navigation

8. Regression-Based CX Driver Analysis Framework

Organization: Telecom Egypt (WE) | **Timeline:** Feb 2021–Present (latest study Jan 2026 – Ongoing capability)

Role: CX Data Analyst (2021) → CX Senior Data Analyst (2022) → CX Data Analytics Manager (2024–Present)

Scale: 5+ Supervised ML Multivariate Linear Regression Studies • **500–75K** Sample Sizes Per Study • **Adjusted R² range: ~0.45–0.99** • **Multiple journey** touchpoints (Digital Chat, Complaint Journeys, Technician Operations, Network/Usage Satisfaction)

Business Problem & Objective

- **Situation:** Leadership needed to understand which operational and experiential factors most significantly impact customer satisfaction metrics (TNPS, RNPS, JNPS). Traditional approaches relied on intuition rather than quantified driver impact, leading to misallocated CX investments and unclear prioritization.
- **Task:** Apply supervised ML regression modeling systematically across multiple CX contexts to quantify driver impact, build predictive frameworks, and enable data-backed prioritization of CX initiatives with measurable ROI.

My Role & Approach

- **Methodology Leadership:** Orchestrated, designed, and executed **5+** regression-based driver analysis studies using **Python, Excel, and Power BI AI Key Influencers**, across FBB, Mobile, Digital, and Field Service journeys; progressed from **IC** hands-on model execution to orchestration and **mentorship/advising** as Manager.
- **Predictive Modeling:** Built linear regression models quantifying the impact of operational KPIs and satisfaction metrics on NPS/JNPS, achieving **Adjusted R²** ranging from **45% to 99.9%** across studies.
- **What-If Predictive Simulation:** Developed scenario simulation frameworks enabling stakeholders to visualize **expected NPS** uplift from specific satisfaction improvements.
- **Statistical Rigor:** Applied multicollinearity detection, coefficient analysis, confidence intervals, and significance testing (e.g. **p-values < 0.001** – where applicable), achieving high explanatory power (**Adjusted R² up to 99.9% for Digital Chat**) to ensure model robustness.
- **Business Translation:** Converted statistical outputs into actionable **KPI targets** and investment prioritization frameworks; facilitated stakeholder negotiations using quantified impact evidence.
- **Key Studies & Findings:**
 - **Touchpoint SAT → RNPS (2021) | FBB I-Use Journey:**
Connectivity ([CONFIDENTIAL] RNPS) and Plan ([CONFIDENTIAL] RNPS) showed the highest uplift potential in scenario simulations → informed prioritization discussions toward high-leverage drivers (e.g. network quality).
 - **Live Chat KPIs → TNPS (2022) | Digital Chat Care:**
FCR ([CONFIDENTIAL]) and Agent SAT ([CONFIDENTIAL]) with **Adjusted R² ≈ 0.99** → enabled data-driven monthly KPI setting for chat operations, establishing FCR and agent engagement as core Live Chat KPIs.
 - **Resolution SAT → JNPS (2024) | FBB I-Complain Journey:**
QA SAT ([CONFIDENTIAL]), Follow-Up Completion ([CONFIDENTIAL]), and Effort ([CONFIDENTIAL]) were dominant drivers; multicollinearity resolved by removing IR variables → clarified that follow-up and resolution quality were far stronger levers than visit satisfaction ([CONFIDENTIAL]x more impactful).
 - **Technician KPIs → JNPS (2024) | FBB I-Complain Journey:**
FCR created [CONFIDENTIAL] percentage-point swing; best vs. worst segments showed [CONFIDENTIAL] percentage-point NPS gap → supported operational threshold setting (e.g., limit repeats/reassignments/complaint volume; ≤[CONFIDENTIAL] complaints, [CONFIDENTIAL] reassigns).

- **Internet SAT → Mobile JNPS (2026) | GSM I-Use Journey:**
GSM Network Quality ([CONFIDENTIAL] JNPS) was the strongest driver; **Adjusted R² ≈ 0.52** → supported converged network CX prioritization (mobile + broadband experience linkage).

Deliverables

- **Regression-based driver quantification models** for 5+ CX journey contexts and interpretation packs (coefficients, significance, and scenario insights).
- **Driver ranking frameworks** to support KPI prioritization and target negotiations.
- **What-If simulation frameworks** enabling scenario-based NPS impact forecasting.
- **Data-backed KPI target recommendations** for operational and satisfaction metrics.
- **Executive presentations and Power BI views** (incl. Key Influencers-style driver explanation) to make insights actionable for non-technical teams.

Key Outcomes & Impact

- **Predictive Accuracy:** Models achieved **Adjusted R²** from **0.45 to 0.99**, explaining substantial variance in customer advocacy.
- **Quantified Driver Impact:** Identified that single KPIs (e.g., FCR) can swing NPS by [CONFIDENTIAL]+ percentage points.
- **Segment-Level Insight:** Revealed [CONFIDENTIAL] percentage-point NPS gap between best and worst operational performance segments.
- **Investment Prioritization Support:** Supported shifting CX investment focus from generic improvements to **high-ROI** drivers (**Connectivity, FCR, QA SAT**).
- **Stakeholder Alignment:** Quantified evidence facilitated negotiations on **KPI targets** and resource allocation.
- **Cross-Journey Application:** Methodology applied across Digital, Field Service, FBB, and Mobile journeys — demonstrating scalability.

Tools & Techniques

Supervised ML (Linear Regression) • Multiple Regression Analysis • What-If Simulation Modeling • Statistical Significance Testing (P-value, t-stat, F-test) • R² / Adjusted R² Validation • Python (statsmodels, scikit-learn) • KNIME • Excel Regression • Power BI (Key Influencer AI Chart) • Variable Selection • Multicollinearity Resolution • Segment Analysis • Stakeholder Presentations

Skills Demonstrated

Supervised ML (Linear Regression) • Predictive Modeling • Statistical Analysis • Driver Quantification • What-If Scenario Modeling • CX Metrics (NPS, TNPS, RNPS, JNPS) • KPI Design & Governance • Data-Driven Prioritization • Executive Stakeholder Communication • Cross-Journey Analytics • Methodology Scalability

9. Detractors Management Program (NPS Closed-Loop Recovery)

Organization: Telecom Egypt (WE) | **Timeline:** May 2024–Present (Ongoing daily operation)

Role: CX Data Analytics Manager (Program Lead & Strategic Advisor)

Scale: ~200K–300K detractors handled to date • [CONFIDENTIAL]– [CONFIDENTIAL] leads/day • [CONFIDENTIAL]– [CONFIDENTIAL] monthly detractor volume • [CONFIDENTIAL] avg attempt rate • [CONFIDENTIAL] avg handle rate • **Cross-functional stakeholders** (Customer Care, CX Insights, Retail, DWH, IT)

Business Problem & Objective

- **Situation:** While the organization tracked NPS and had a systematic closed-loop process to engage detractors, leadership aimed to further enhance the system to capture more actionable customer feedback. NPS improvement primarily relied on immediate service recovery with limited focus on structured indirect and long-term initiatives.
- **Task:** Lead the design and implementation of an enterprise-wide Detractors Management Program—enhancing handling processes, aligning stakeholders, building reporting frameworks, and planning long-term improvement projects—to reduce detractor volume and enhance NPS through direct customer engagement and systemic fixes.

My Role & Approach

- **Program Leadership:** Managed and led analytics team to perform required analysis; designed enhancements to the existing detractor handling process and negotiated implementation with Customer Care stakeholders.
- **Process Enhancement:** Proposed and secured agreement on enhanced handling process including [CONFIDENTIAL] detractor/day handling target with [CONFIDENTIAL] retail allocation.
- **Stakeholder Negotiation:** Navigated cross-functional discussions with Customer Care, CX Insights, Retail, DWH, and IT teams to align on process, data requirements, and reporting needs.
- **Dashboard Design:** Specified dashboard requirements for tracking detractor handling process outcome, attempt rates, handle rates, and recovery outcomes; guided team on Power BI implementation.
- **Pilot Validation:** Conducted [CONFIDENTIAL]-detractor pilot to measure handling effectiveness and validate recovery potential before scale-up.
- **Phased Delivery:**
 - **Milestone 1:** Start Handling I-Model IVR Detractors ✓ ([CONFIDENTIAL]/day achieved)
 - **Milestone 2:** Enhance Handling Process ✓ (Process improvements implemented)
 - **Milestone 3:** Design Detractors Post-Survey ✓ (Survey designed and recorded)
 - **Milestone 4:** Dashboard & Reporting Enhancement (In Progress)
 - **Milestone 5:** Detractors-Based Improvement Projects (Not Started)

Deliverables

- **New stakeholder agreement** defining detractor handling process, responsibilities, and targets.
- **End-to-end process** definition documentation for detractor management.
- **Detractor handling dashboard** (Power BI) tracking attempt rates, handling rate, and recovery outcomes.
- **Monthly reporting framework** for detractor insights, trends, and executive review.
- **Post-survey instrument** for handled detractor feedback (recorded; pending launch).

Key Outcomes & Impact

- **Pilot Recovery Validation:** [CONFIDENTIAL]-detractor pilot demonstrated strong recovery: **61.4%** converted to Promoters, [CONFIDENTIAL] to Passives, only [CONFIDENTIAL] remained Detractors.
- **Operational Scale:** ~200K–300K I-Model IVR detractors handled to date through systematic closed-loop process ([CONFIDENTIAL]– [CONFIDENTIAL]/day × 22 months)
- **Average Performance:** [CONFIDENTIAL] attempt rate and [CONFIDENTIAL] handle rate against [CONFIDENTIAL]– [CONFIDENTIAL] monthly detractor volume.
- **Peak Performance:** Achieved [CONFIDENTIAL] Retail attempt rate, [CONFIDENTIAL] handle rate, and [CONFIDENTIAL] total attempt rate and [CONFIDENTIAL] total handle rate (July 2025 — post new agreement; headcount constraints limited sustainability)
- **Channel Insight:** Retail channels consistently outperform Non-Retail in closing the loop, informing resource allocation.
- **Cross-Functional Alignment:** Secured unified framework agreement across Customer Care, CX Insights, Retail, DWH, and IT.
- **Status:** M1–M3 complete; M4 (dashboard enhancement) in progress; M5 (detractor-based projects) pending initiation.

Tools & Techniques

Process Design & Enhancement • Stakeholder Negotiation • Dashboard Requirements Specification • Power BI • Excel • Pilot Testing • Recovery Analytics • Cross-Functional Orchestration • Closed-Loop Feedback Design

Skills Demonstrated

Program Leadership • Stakeholder Management • Process Design & Improvement • Cross-Functional Negotiation • Dashboard Specification • Pilot Design & Validation • Recovery Analytics • Strategic Roadmap Planning • Top Management Recommendations • Channel Performance Analysis

10. CX Dashboards & KPI Governance (Dynamic Dashboards + Executive Reporting)

Organization: Telecom Egypt (WE) | **Timeline:** 2019–Present (Ongoing capability)

Role: (From IC to Team Lead/Advisor) CX Data Analyst (2019–2022) → CX Senior Data Analyst (2022–2023) → CX Data Analytics Manager (2023–Present)

Scale: 30+ interactive dashboards • 100+ static reports • 100+ KPIs designed • 12+ departments served • 42% report turnaround improvement • 15+ ad-hoc analyses/month

Business Problem & Objective

- **Situation:** The organization had no centralized CX dashboards and relied on manual Excel/PowerPoint reports, leading to slow turnaround, inconsistent metrics, and limited executive visibility into customer experience performance across channels.
- **Task:** Architect a comprehensive suite of dynamic Power BI dashboards and executive reporting frameworks. Build an enterprise-wide CX analytics and reporting capability—establishing interactive dashboards, standardized KPI frameworks, and automated reporting pipelines—to empower data-driven decision-making across 12+ departments.

My Role & Approach

- **Capability Building:** Progressed from hands-on IC building the first CX dashboards (2019) to leading and orchestrating dashboard strategy as Manager (2023–Present).
- **Requirements & Stakeholder Alignment:** Collaborated with executives and stakeholders across 12+ departments to define business objectives, research questions, and KPI definitions.
- **Data Engineering:** Designed data collection processes, performed data wrangling, and applied feature engineering to produce high-quality analytical datasets.
- **KPI Governance:** Managed the design and rollout of 100+ cross-channel KPIs (e.g., NPS, TNPS, CSAT, CES, AHT, FCR, IVR containment, Footfall, Conversion Rate) to standardize performance measurement.
- **Dashboard Development:** Built 30+ interactive Power BI dashboards covering Digital, Retail, Contact Center, Technician Operations, and Enterprise segments.
- **Process Automation:** Engineered automated data pipelines using Power Query (M Language) and DAX, replacing most of manual wrangling and achieving 42% reduction in report turnaround time.
- **Reporting Excellence:** Delivered 100+ static reports (Daily, Weekly, Monthly, Quarterly, Ad-Hoc) with actionable insights and recommendations for top management.
- **Prototype Leadership:** Dashboards and design standards served as prototype and design guide for IT DWH BI Team's enterprise-wide reporting.
- **Dashboard Portfolio (Selected Examples):**
 - **CX Surveys & NPS:** Digital Survey Dashboard (1st CX Dashboard) | Journey-Based Dashboard, MSAN TNPS Dashboard | 2019–2020 | IC Role
 - **Operations & Process:** Outage Dashboard, TTS Logs Process Mining, Installation New Process POC | 2020 | IC Role
 - **Commercial & Lifecycle:** GSM MNP Dashboard (Port-IN/OUT), WE Space Dashboard | 2019–2020 | IC
 - **Enterprise Experience:** SMEs Experience Scan V1 & V2, Technical Support Phase, Corporate GSM, Compound Experience | 2022–2024 | Lead/Advisor Role

- **Integrated with Projects:** Personas, Churn, Culture Assessment, Platinum Segment, Driver Analysis, Detractors | 2019–Present | Role Varies

Deliverables

- **30+ interactive Power BI dashboards** for executive and operational decision-making.
- **100+ static reports** (Daily, Weekly, Monthly, Quarterly, Ad-Hoc) with insights and recommendations.
- **100+ KPI definitions** and governance frameworks across CX, Digital, Retail, and Contact Center.
- **Automated data pipelines** (Power Query/DAX) reducing manual effort by 90%.
- **Dashboard design standards** adopted as prototype by IT DWH BI Team.

Key Outcomes & Impact

- **Enterprise Coverage:** 30+ dashboards serving executives and stakeholders across 12+ departments.
- **KPI Standardization:** 100+ cross-channel KPIs defined and governed (NPS, TNPS, CSAT, AHT, FCR, IVR, Footfall, Conversion, etc.).
- **Process Efficiency:** 42% reduction in report turnaround time through automated pipelines.
- **Operational Responsiveness:** 15+ ad-hoc analyses delivered monthly to support time-sensitive decisions.
- **Prototype Recognition:** Dashboard designs and standards adopted by IT DWH BI Team as enterprise-wide prototype.
- **Foundation for Projects:** Dashboards powered decision-making for Personas, Churn, Culture, Platinum, Driver Analysis, and Detractors projects.
- **Role Progression:** Evolved from IC (building first CX dashboard) to leading enterprise reporting strategy.

Tools & Techniques

Power BI (DAX, Power Query/M, Data Modeling, ERD) • SQL • Excel Power Query • Microsoft Access • PowerPoint • KPI Design & Governance • Automated Data Pipelines • EDA • Feature Engineering • Statistical Validation • Executive Reporting

Skills Demonstrated

Dashboard Development • KPI Design & Governance • Data Visualization • Data Modeling • Process Automation • EDA • Feature Engineering • Executive Reporting • Stakeholder Management • Requirements Engineering • Prototype Design • Scalable Analytics Architecture • IC-to-Leadership Progression

11. VOC Sentiment Analysis (Arabic/English Daily Text Analytics)

Organization: Telecom Egypt (WE) | **Timeline:** 2022 (Completed)

Role: CX Senior Data Analyst

Scale: 10K daily comments • 9K+ keyword/search-item codebook • **Bilingual** (Arabic/English) • **Multi-channel** (Call Center, Chat, Retail, Social) • **97%** time-reduction • Daily output generated in **~4 minutes** (vs ~2 hours manual) • **70%+** accuracy

Business Problem & Objective

- **Situation:** The organization's VOC analysis ran monthly on an automated basis, but lacked daily real-time capability to detect and respond to emerging customer issues. TNPS drops and spikes could not be explained quickly, delaying root cause identification and operational response.
- **Task:** Build a daily VOC analysis capability to capture FBB customer sentiment on-the-spot, enabling same-day detection of issues and rapid root cause analysis for TNPS fluctuations.

My Role & Approach

- **Codebook Development:** Built comprehensive 9K-keyword search codebook (building on teammate's initial foundation) covering Arabic and English customer expressions and themes.
- **Algorithm Design:** Developed rule-based text classification algorithm using keyword matching to categorize customer verbatims into actionable themes.
- **Data Engineering:** Wrangled TNPS survey open-text data from multiple channels (Call Center, Chat, Retail, Social) into analysis-ready structure using Excel Power Query (M Language).
- **Visualization:** Created simple, clear Excel view to enable CX team to capture VOC trends easily and quickly.
- **Validation:** Achieved 70%+ accuracy validated through manual random sample checking.

Deliverables

- **9K-keyword** bilingual (Arabic/English) codebook for VOC classification.
- **Rule-based VOC** analysis algorithm in Excel Power Query (M Language).
- **Daily VOC** trend view for CX team consumption.
- **Automated process** reducing analysis time from 2 hours to 4 minutes.

Key Outcomes & Impact

- **Time Efficiency:** Reduced daily VOC analysis from 2 hours to 4 minutes — 97% reduction in processing time.
- **Accuracy:** Achieved 70%+ classification accuracy (validated via manual random sample check)
- **Volume Processed:** 10K+ daily comments analyzed across multiple channels.
- **Early Detection:** Enabled same-day identification of emerging issues (e.g., Quota Consumption patterns).
- **Actionability:** Findings drove operational and strategic responses to VOC-detected issues.
- **Bilingual Capability:** Processed both Arabic and English customer verbatims.

Note: Project completed in 2022; tool was operational for FBB VOC analysis during that period.

Tools & Techniques

Excel Power Query (M Language) • Rule-Based Text Classification • Keyword Codebook Development • VOC Analysis • Bilingual Text Processing (Arabic/English) • Manual Validation Sampling • Data Wrangling

Skills Demonstrated

Text Analytics • VOC/Verbatim Analysis • Codebook Development • Rule-Based Classification • Process Automation • Bilingual (Arabic/English) Text Processing • Data Wrangling • Rapid Prototyping • Root Cause Enablement • TNPS Analysis

12. CX Field Research & Founding Team Contributions

Organization: Telecom Egypt (WE) | **Timeline:** 2018–2022 (Founding CX Team Member)

Role: CX Data Analyst (Founding Team Contributor)

Scale: **81+** branch visits • **12** governorates • **Face-to-face** surveys at **2 major** venues (WiFi venue customers at Semoha & Heliopolis Clubs) • **FTTH** survey campaign support • Introduced **iAuditor** initiative (paperless mystery shopping) • **McKinsey SAHM** Journey Workshop (2022) participation as personas creator/presenter • **Manual VOC** coding support

Business Problem & Objective

- **Situation** As a founding member of the CX team, the organization needed to build foundational capabilities beyond data analytics—including physical touchpoint evaluation, direct customer feedback collection, and cross-functional CX advocacy. These field activities were essential to ground data-driven insights in real-world customer experiences.
- **Task:** Contribute to, help shape and establish a wide range of CX foundational field research capabilities, includes mystery shopping, survey execution, process innovation, VOC support, and event planning—to build organizational CX maturity and complement analytical work with hands-on customer understanding.

My Role & Approach

- **Mystery Shopping Execution:** Conducted **81+** branch visits across **12** governorates to assess service standards, staff behavior, and journey compliance using structured assessment protocols.
- **Face-to-Face Surveys:** Executed surveys with WiFi venue customers at Semoha and Heliopolis Clubs to capture in-the-moment experience feedback.
- **FTTH Campaign Support:** Coordinated survey logistics and execution for Fiber-to-the-Home customer feedback collection.
- **Manual VOC Coding:** Provided coding support for open-ended verbatims to help structure unstructured feedback (foundation for automated VOC sentiment analysis built in 2022).
- **Process Digitization Initiative:** First to pilot iAuditor mobile app to **replace paper-based mystery shopping forms**, improving data capture speed and enabling mobile-first field research.
- **Vendor Engagement:** Initiated contact and negotiation with **Checker Soft** for mystery shopping platform before handover to CX Planning team.
- **Cross-Functional Collaboration:** Co-planned and executed CX orientation event; presented customer personas (2019) at McKinsey SAHM Customer Journey Workshop (2022) to cross-departmental participants.

Deliverables

- **81+ mystery shopping assessment** reports across **12 governorates** (retail branches).
- **Face-to-face** survey datasets from WiFi venue customers (**2 clubs**).
- **FTTH survey** campaign execution (**logistics and fieldwork coordination**).
- **Manual VOC coding** input for unstructured feedback structuring.
- **iAuditor** app pilot documentation and workflow recommendations.
- **Checker Soft vendor engagement** documentation (RFI, negotiation, handover).
- **CX orientation event** (planning and execution).
- **Personas workshop presentation** (McKinsey SAHM strategic project, 2022).

Key Outcomes & Impact

- **Field Coverage:** 81+ branch visits spanning 12 governorates established baseline service quality benchmarks for retail touchpoints.
- **Process Innovation:** iAuditor app pilot replaced paper-based mystery shopping workflows, enabling mobile-first data capture (initiative handed to CX Planning for scale).
- **Vendor Enablement:** Checker Soft engagement streamlined vendor onboarding process for CX Planning team.
- **Strategic Integration:** Presented personas to McKinsey and WE department leads during SAHM workshop (2022), embedding personas into enterprise transformation roadmap (adopted in 2023 enterprise-scale personas refresh).
- **VOC Program Foundation:** Manual coding and field survey work contributed to later development of automated VOC sentiment analysis (2022) and structured feedback frameworks.
- **Multi-Channel VOC Feeders:** Established foundational field research capabilities (mystery shopping, surveys, manual coding) feeding into enterprise VOC program.

Note: This project focused on enablement and foundational CX program build-out; direct ROI/NPS impact not isolated to these activities alone.

Tools & Techniques

Field Research Protocols • Mystery Shopping Assessment • Face-to-Face Survey Administration • iAuditor Mobile App (Pilot) • Manual VOC Coding • Vendor Management (RFI/Negotiation) • Workshop Facilitation • Cross-Functional Stakeholder Engagement

Skills Demonstrated

Field/Retail CX Measurement • VOC Program Enablement • Process Digitization • Vendor Management • Manual Text Coding • Survey Design & Execution • Cross-Functional Collaboration • Strategic Stakeholder Engagement • Initiative & Ownership • CX Program Build-Out



Applied Data Science & Machine Learning Portfolio

13. Data Science & Machine Learning Portfolio (Self-Directed Learning)

Organization: Self-Directed (Kaggle, Udacity/FWD, Udacity/Bertelsmann) | **Timeline:** 2020–2021

Role: Independent Learner & Practitioner

Portfolio (GitHub): [View Data Science Repository](#)

Scale (Verified): 12 end-to-end projects | 5 Regression | 1 Classification | 1 Clustering/PCA | 1 Time Series | 1 A/B Testing | 2 Data Wrangling | 1 Capstone (Multi-Method)| Largest dataset: 500K records | Kaggle Top 30% (8,040 participants; 60,288 submissions)

Tools: Python (Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn, tweepy, requests), Alteryx, Tableau, Excel

Portfolio Summary

Metric	Value
Total Projects	12 end-to-end
Method Applications	15 (Capstone spans 3 disciplines)
Total Records Processed	500K+
Kaggle Ranking	Top 30% (8,040 participants; 60,288 submissions)
Models Built & Compared	25+
ML Disciplines	6 (Regression, Classification, Clustering, Time Series, A/B Testing, Data Wrangling)
Tool Stacks	Python, Alteryx, Tableau, Excel

Strategic Highlight

Built a technical sandbox to master the **full-stack data science lifecycle**—from API ingestion to predictive modeling to business recommendation. Directly transferred these capabilities to Telecom Egypt enterprise projects: customer personas (22M+ customers), churn prediction (74%+ recall), and CX driver analysis (R^2 0.45–0.99) .

Business Problem & Objective

- **Situation:** To strengthen my hands-on data science execution beyond corporate CX analytics, reporting, and dashboards, I completed an intensive self-directed curriculum of real-world projects covering the full data science lifecycle—from API data extraction and complex data wrangling to predictive modeling (supervised and unsupervised learning), forecasting, experimentation, statistical validation, and business insight translation.
- **Task:** Execute 12 end-to-end projects demonstrating hands-on proficiency across supervised learning, unsupervised learning, time series forecasting, experimentation, and data engineering—using industry-standard tools and methodologies with clear decision framing, validated models, and business-ready recommendations.

My Role & Approach

- **End-to-End Execution Model:** For each project, independently owned the complete pipeline:
Problem framing, hypothesis definition, & decision criteria → Data acquisition (files/APIs) → Data quality assessment, Cleaning/wrangling → Exploratory analysis (EDA) → Feature engineering → Model development, validation, evaluation & benchmarking → Insights communication & business recommendation.

Technical Execution: Projects by Discipline

1. Supervised Learning: Regression (5 Projects)

- **Business Problem:** Various business decisions requiring continuous value predictions—auction bidding valuation, marketing ROI estimation, cost prediction, and retail location expansion.

- **Use Case:** Predict insurance claim amounts for risk assessment; recommend diamond auction bid price (\$8.2M); estimate catalog mailing profit (\$21,987); prepare data for per-pupil cost prediction; recommend best city for pet store expansion.
- **Model/Algorithm:** Built predictive models using Linear Regression (with ordinal and dummy variable encoding), XGBoost, and LightGBM.

Project	Scale	Model	Key Results	Link
Insurance Claims (Kaggle)	300K train / 200K test	XGBoost, LightGBM	Top 30% of 8,040; RMSE 0.725; 2-fold CV	View
Diamond Pricing	5K train / 3K predict	MLR (Dummy)	R ² 0.92; \$8.2M bid recommendation (70%)	View
Catalog Demand	2.3K train / 250 predict	MLR	R ² 0.84; recommendation: send catalog (profit > \$10K threshold) \$21,987 profit	View
School Cost Prediction	Messy dataset	Data Prep	Regression-ready; validated predictors (correlation/R ² /P-values)	View
Pet Store Sales	11→10 cities	Outlier Analysis	Best predictor R ² 0.82; location recommendation	View

- **Selection Criteria:** Adjusted R² for explanatory power; P-values for predictor significance; RMSE for prediction accuracy; Kaggle leaderboard ranking for competitive validation.
- **Validation Methods:** Cross-validation (K-Fold), train/test split, R² evaluation, feature importance analysis, hyperparameter tuning, residual analysis.
- **Key Insight:** Dummy variable encoding (R² 0.92) outperformed ordinal encoding (R² 0.90) in Diamond Pricing—categorical treatment of cut/color/clarity captured non-linear relationships better than assumed ordinal hierarchy.

2. Supervised Learning: Classification (1 Project)

- **Business Problem:** Small bank experienced sudden influx of 500 loan applications due to competitor scandal; needed to identify creditworthy customers within one week.
- **Use Case:** Produced creditworthy customer list from 500 loan applicants (408 approved, 92 rejected).
- **Model/Algorithm:** Developed and benchmarked 4 binary classification algorithms: Logistic Regression, Decision Tree, Random Forest, Gradient Boosted Model.

Project	Scale	Models Compared	Best Model	Key Results	Link
Credit Default Risk	500 applicants	4 models compared (Logistic Regression, Decision Tree, Random Forest, GBM)	Random Forest	79.3% accuracy, 73.7% AUC, 408 creditworthy identified	View

- **Selection Criteria:** Best overall accuracy + largest AUC-ROC + acceptable prediction bias (6.5%).
- **Validation Methods:** 70/30 train/validation split; Accuracy, Precision/Recall per class, AUC-ROC comparison with documented performance trade-offs.
- **Key Insight:** Random Forest (79.3% accuracy, 73.7% AUC) outperformed Boosted Model despite similar accuracy—lower bias and better precision balance made it more reliable for business decision.
- **Capstone Application:** Applied same multi-model comparison approach in Capstone—Boosted Model selected (76% accuracy, 83% F1) for classifying 10 new stores into optimal formats after benchmarking against Decision Tree and Random Forest.

3. Unsupervised Learning: Clustering (1 Project)

- **Business Problem:** Retail chain seeking international expansion needed to identify countries economically, demographically, educationally, and environmentally similar to USA.
- **Use Case:** Identified USA-similar countries for market expansion prioritization using multi-dimensional similarity analysis.
- **Model/Algorithm:** Applied PCA for dimensionality reduction followed by clustering comparison: K-Means, K-Medians, Neural Gas.

Project	Techniques	Validation	Key Results	Link
Country Segmentation	PCA + Neural Gas	Rand Index, Calinski-Harabasz (CH) Index, Tableau Map	3 clusters; USA-similar countries identified for market expansion	View

- **Selection Criteria:** Rand Index (cluster agreement), Calinski-Harabasz Index (cluster separation), visual validation via Tableau geographic map.
- **Validation Methods:** Statistical cluster validation (Rand/CH Index), geographic sense-check, business logic alignment.
- **Key Insight:** Neural Gas outperformed K-Means and K-Medians on CH Index at k=3—topology-preserving properties better handled the multi-dimensional country similarity space.
- **Capstone Application:** Applied same 3-algorithm comparison in Capstone—K-Means selected (3 clusters, Rand/CH validated) for segmenting 85 stores into optimal formats. Cluster outputs became classification labels for downstream modeling.

4. Time Series Forecasting (1 Project)

- **Business Problem:** Video game supply chain needed 4-month sales forecast to plan inventory and coordinate with distributors.
- **Use Case:** Forecasted monthly video game sales for next 4 periods with 80% and 95% confidence intervals for supply chain planning.
- **Methodology:**

Step	Technique	Purpose
1	Decomposition Plot	Identify Error (M), Trend (A/N), Seasonality (M)
2	ACF/PACF Analysis	Determine differencing (d, D) and AR/MA terms
3	Model Building	Configure ETS and ARIMA parameters
4	In-Sample Evaluation	RMSE, MASE, MAPE comparison
5	Holdout Validation	Test predictive accuracy on unseen data
6	Forecast Generation	80% and 95% confidence intervals

- **Model/Algorithm:** Built and compared ETS models (M,A,M and M,Ad,M with damping) and ARIMA(0,1,1)(0,1,0)[12] using decomposition analysis and ACF/PACF diagnostics.

Project	Models Compared	Final Model	Key Results	Link
Video Game Forecasting	ETS(M,A,M), ETS(M,Ad,M), ARIMA(0,1,1)(0,1,0)[12]	ARIMA	RMSE ~37K, MASE 0.36; 4-month forecast	View

- **Selection Criteria:** Holdout sample performance prioritized over in-sample fit; RMSE, MASE (<1.0 threshold), MAPE comparison.
- **Validation Methods:** 4-period holdout validation, decomposition analysis (Error/Trend/Seasonality), ACF/PACF for differencing strategy and AR/MA term identification.
- **Key Insight:** ARIMA selected despite worse in-sample fit (RMSE ~37K vs ~31K)—holdout validation revealed better generalization, demonstrating that validation performance is more reliable than training metrics.
- **Capstone Application:** Applied same ETS vs ARIMA comparison in Capstone—ETS(M,N,M) selected (MASE 0.44) for produce sales forecasting using 6-month holdout validation. Opposite model won, confirming importance of per-dataset evaluation.

5. A/B Testing & Experimentation (1 Project)

- **Business Problem:** Coffee shop chain considering new menu launch with TV advertising; needed to determine if expected profit justifies marketing investment (18% lift threshold).
- **Use Case:** Designed and analyzed matched-pair experiment to recommend menu rollout decision with quantified expected impact (\$6.9M).
- **Methodology:**

Aspect	Details
Design	Matched-Pair A/B Test
Treatment	10 stores with new menu + TV advertising
Control	20 stores (2 matched per treatment)
Test Period	12 weeks
Metric	Weekly Gross Margin

- **Model/Algorithm:** Matched-Pair A/B Test design with 4 control variables (Trend, Seasonality, AvgMonthSales, Region); multicollinearity handling by selecting highest-correlation predictor.\

Project	Design	Test Period	Key Results	Link
Coffee Shop Menu Launch	Matched-Pair (10:20)	12 weeks	41% lift, P=0.000, \$6.9M expected	View

- **Control Variable Selection:**

Variable	Correlation	Included	Reason
AvgMonthSales	0.999	Yes	Highest correlation
Region	Strong	Yes	Geographic control
Trend	-	Yes	Time-based pattern
Seasonality	-	Yes	Cyclical pattern
AvgMonthTrafficCount	Strong	No	Multicollinear with Sales
Sq_Ft	Weak	No	No correlation

- **Regional Results:**

Region	Lift	P-Value	Confidence	Expected Lift
Western	36.3%	0.006%	99.4%	\$4,879,403
Central	45.8%	0.004%	99.6%	\$2,019,039
Overall	41.0%	0.000%	100%	\$6,898,441

- **Selection Criteria:** Practical significance (≥18% lift threshold) AND statistical significance (P < 0.05).

- **Validation Methods:** T-test for significance, lift calculation by region, P-value evaluation, treatment-control matching validation.
- **Key Insight:** Central region showed higher lift (45.8%) than Western (36.3%) despite Western contributing more absolute value (\$4.9M vs \$2.0M)—regional breakdown revealed implementation priority opportunities beyond aggregate results.
- **Recommendation:** Roll out new menu to all stores — exceeded 18% practical significance threshold with 41% lift.

6. Data Wrangling & Engineering (2 Projects)

- **Business Problem:** WeRateDogs: Analyze Twitter account engagement patterns from fragmented multi-source data. Bikeshare: Enable interactive exploration of bikeshare usage patterns across 3 US cities.
- **Use Case:** WeRateDogs: Answered 6 research questions about dog ratings and engagement after resolving 30 data quality issues. Bikeshare: Built interactive CLI tool for on-demand descriptive statistics.
- **Model/Algorithm:** WeRateDogs: Multi-source data pipeline (tweepy API, HTTP requests, regex extraction, pandas merging/melting). Bikeshare: Modular Python functions (9 total) with datetime parsing, groupby operations, user input validation.

Project	Data Sources	Techniques	Key Results	Link
Twitter WeRateDogs	3 (CSV, TSV, Twitter API)	tweepy, requests, regex	30 issues resolved (22 quality, 6 tidiness, 2 feature engineering); 6 research questions; 2 reports	View
US Bikeshare Explorer	3 cities (CSV)	Pandas, Interactive CLI	9 modular functions; 3 cities; 5 statistics categories	View

- **Selection Criteria:** Data completeness, consistency, tidiness (one variable per column, one observation per row), analytical readiness.
- **Validation Methods:** Programmatic quality assessment, visual inspection, before/after comparison, formal wrangling documentation (2 reports).
- **Key Insight:** WeRateDogs regex extraction initially failed for edge cases (decimal ratings, aggregated scores)—iterative pattern refinement and manual validation resolved 22 quality issues that automated detection missed.
- **Output:** Production-ready analytical datasets; 2 formal wrangling reports documenting methodology and cleansing decisions.

7. Capstone: Multi-Method End-to-End Pipeline (1 Project / 3 Disciplines)

- **Business Problem:** Grocery chain with 85 stores suffering product surpluses and shortages from treating all stores identically; opening 10 new stores without format guidance.
- **Use Case:** End-to-end solution: Segmented 85 stores into 3 optimal formats → Classified 10 new stores into best-fit format → Forecasted produce sales for inventory planning.
- **Model/Algorithm:** Clustering: K-Means, K-Medians, Neural Gas. Classification: Decision Tree, Random Forest, Boosted Model. Time Series: ETS (3 variants), ARIMA (3 variants). Total: 12 models benchmarked.

Task	Method	Key Outcome
Store Segmentation	K-Means (3 clusters)	85 stores → 3 formats (25/35/25)
New Store Classification	Boosted Model	76% acc, 83% F1, 10 stores assigned
Sales Forecasting	ETS(M,N,M)	MASE 0.44, 80%/95% CI

- **Top Predictors:** Age0to9, Hval750Kplus, Age65Plus
- **Selection Criteria:** Clustering: Rand Index, CH Index at k=3. Classification: Overall accuracy, F1 Score, error rate, precision/recall balance. Time Series: Holdout MASE, RMSE; AIC for model complexity.
- **Validation Methods:** Clustering: Box-whisker diagnostics, CH/Rand Index. Classification: 20% holdout (Random Seed = 3). Time Series: 6-month holdout, ETS vs ARIMA comparison.
- **Key Insight:** End-to-end pipeline integration—clustering output (store formats) became classification labels, which then determined forecast groupings. Upstream model decisions cascade through the entire analytical workflow.
- **Link:** [View Capstone Project](#)

Deliverables

- **12 end-to-end project solutions** with documented code, methodology, and business recommendations
- **Reusable data pipelines:** Handling CSV, JSON, TXT, and API extraction (Twitter)
- **Formal reports:** Methodology documentation, assumptions, limitations, recommendations

Key Outcomes & Impact

Outcome	Evidence
Kaggle Competitive Validation	Top 30% (outperforming ~70% of 8,040 participants 7,573 Teams 60,288 Submissions) in Insurance Prediction; RMSE 0.725 using LightGBM with K-Fold CV on 500K total records
Multi-Model Benchmarking Framework	Systematic comparison of 25+ models across 12 projects spanning 6 ML disciplines with documented selection rationale
Technical Breadth	Hands-on proficiency across 6 ML disciplines (regression, classification, clustering, time series, A/B testing, data wrangling) using 4 tool stacks
Data Engineering Rigor	Resolved 30 data quality issues in single project (22 quality, 6 tidiness, 2 feature engineering); gathered data from 3 sources (CSV, TSV, Twitter API via tweepy); delivered 2 formal reports documenting methodology and analysis conclusions.
Statistical Rigor	Applied R^2 , AUC-ROC, cross-validation, confidence intervals, and hypothesis testing
Business Translation	Each project framed as business decision (profit estimation, bidding strategy, loan approval, demand planning, go/no-go)
Corporate Transfer	Techniques directly applied at Telecom Egypt: clustering → personas (22M+), regression → CX drivers (R^2 0.45–0.99), Time series → Demand forecasting concepts, A/B testing → experiment design

Tools & Techniques

- **Programming:**
Python (Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn, tweepy, requests) • Jupyter Notebook
- **Analytics Platforms:**
Alteryx Designer (KNIME Transferable-Capability) • Tableau • Excel
- **ML Methods:**
Linear Regression • Logistic Regression • Decision Tree • Random Forest • Boosted Models • K-Means • K-Medians • Neural Gas • PCA • ETS • ARIMA
- **Experimentation:**
A/B Testing • Matched-Pair Design • Hypothesis Testing • Confidence Intervals

- **Data Engineering:**

API Extraction • Multi-Format Processing (CSV, JSON, TXT) • Data Quality Assessment • Feature Engineering

Skills Demonstrated

Supervised ML (Regression/Classification) • Unsupervised ML (Clustering/PCA) • Time Series Forecasting (ETS/ARIMA) • Time Series Decomposition • ACF/PACF Analysis • Seasonal Differencing Strategy • Dampened Trend Modeling • A/B Testing & Experimentation • Matched-Pair Design • Treatment-Control Matching • Multi-Model Benchmarking (25+ models) • Holdout Validation Design • Data Wrangling • API Integration • Feature Engineering • Statistical Validation • Statistical & Practical Significance • Holdout-Based Model Selection • Kaggle Competition • Business Problem Translation • End-to-End Project Ownership

Bottom Line: *This portfolio demonstrates that my corporate CX analytics leadership is built on a foundation of hands-on, end-to-end data science execution and applied machine learning capability — not just tool knowledge, but proven modeling, validation, and business translation capability.*



Leadership, Training & Capability Building

14. Leadership & Data Capability Building

Organization: Independent Self-Employed (Physical/Remote) | **Timeline:** Jan 2021 – Present

Role: Freelance Data Science & CX Data Analytics Trainer | **Training Portfolio (GitHub):** [View Training Repository](#)

Scope: Data Science, Analytics, Power BI, and CX Analytics enablement for professionals and early-career analysts.

Scale: 30+ Programs • 540+ Instruction Hours • 250+ Professionals Trained • 98% Satisfaction

Target Audience: Enterprise Leaders, Data Professionals, and Early-Career Analysts

Clients: Telecom Egypt (CX Data Analytics Corporate Training), [ProPath Academy](#) (Saudi Arabia), Professional Network

Training Impact Summary

Metric	Value
Programs Delivered	30+
Training Hours	540+
Professionals Trained	250+
Satisfaction Rating	98%
Curricula Developed	10+ end-to-end
Global Reach	2 Markets – Egypt & Saudi Arabia (Physical/Remote)

Strategic Highlight

To democratize data literacy by **bridging the gap between theoretical data science and commercial application**. I specialize in transforming raw technical concepts into "Decision Intelligence" for business leaders and frontline analysts—grounded in 10+ years of enterprise CX analytics experience.

Programs emphasize end-to-end analytical thinking: from problem framing and data preparation to model interpretation and business recommendation.

Business Problem & Objective

- **Situation:** In many emerging analytics markets, there is a gap between academic data science theory and real-world commercial application — particularly in telecom and CX analytics contexts. Professionals often learn tools but struggle to translate outputs into business decisions.
- **Task:** Design and deliver practical, industry-oriented, and end-to-end training programs that bridge this gap—equipping professionals across corporate and external training channels with the analytical mindset and technical skills required to apply data science in real operational contexts.

My Role & Approach

- **Curriculum Development:**
 - Independently designed 10+ end-to-end curricula aligned with enterprise needs and industry certifications (PL-300 standards for Power BI).
 - Structured every program around business outcomes—each ends with a deliverable (dashboard, model comparison report, or structured analysis).
- **Delivery Channels:**

Channel	Scope
Telecom Egypt (On-Job)	Internal capability building for CX data analysts
Telecom Egypt (Official Events)	Corporate CX training programs
ProPath Academy (Saudi Arabia)	Power BI & Data Analysis Course (36 graduates)
Professional Network	Custom coaching and skill workshops

- **Training Methodology:**
 - **Outcome-Driven:** Every program produces a tangible deliverable aligned with business needs.
 - **Business Translation:** Teaches "how to decide" (selection criteria, validation, trade-offs)—not just "how to build".
 - **Adaptive Delivery:** Format (physical/remote) and depth adjusted to learner profile—from entry-level analysts to practicing managers.
 - **Feedback-Integrated:** Projects, quizzes, and review sessions embedded throughout.

Curriculum Portfolio

Domain	Topics Covered
Data Science Foundations	Python (Pandas, NumPy), SQL, Statistics, Probability, Machine Learning (Regression, Classification, Clustering)
Analytics & Visualization	Power BI (DAX, Power Query, Data Modeling), Tableau, Excel (Power Query, Power Pivot), Alteryx, KNIME
CX Analytics	Journey Mapping, Persona Development, VoC/VoE Analytics, NPS/TNPS, Driver Analysis, KPI Design
Methodologies	CRISP-DM, Analytical Thinking, Data Analysis Process, EDA, Feature Engineering, Model Validation, Statistical Validation, Hypothesis Testing

Deliverables

- **10+** end-to-end curricula (Python, SQL, Power BI, ML, CX Analytics, CRISP-DM).
- **Hands-on labs** with real datasets and capstone projects.
- **Formal training materials**, assessments, and feedback frameworks.
- **Remote delivery infrastructure** for cross-border programs.

Key Outcomes & Impact

- **High-Stakes Delivery:** 98% satisfaction across **540+** hours of instruction.
- **Enterprise Adoption:** Delivered recurring training sessions for Telecom Egypt internal cohorts (on-the-job + official events)
- **Capability Transformation:** Delivered training that enabled upskilling colleagues from manual reporting to automated Power BI governance.
- **Cross-Border Scale:** Delivered specialized certification paths for ProPath (Saudi Arabia)—**36** analysts graduated.
- **Certification Alignment:** Curriculum designed to PL-300 (Microsoft Power BI Data Analyst) standards.

Training Delivery Platforms

Google Meet • Microsoft Teams • Zoom

Tools & Techniques

- **Methodologies:** CRISP-DM • Statistical Validation • Hypothesis Testing • Analytical Thinking • CX Framework
- **Programming:** Python • Jupyter Notebook • SQL
- **Analytics Platforms:** Microsoft Power BI (Data Model, ERD, DAX, Power Query) • Tableau • Excel • Alteryx Designer • KNIME

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Skills Demonstrated

Curriculum Design • Technical Communication • Adult Learning Principles • Enterprise Capability Building • Cross-Functional Stakeholder Training • Remote & Physical Delivery • Program Management • Business Translation

Differentiator

Training experience demonstrates ability to translate complex technical concepts into actionable business skills—bridging the "knowing vs doing" gap that limits most analytics professionals. A key differentiator for leadership roles requiring team development and organizational capability building.